Feasibility Memo - Week 5

Executive Verdict

The project to build a predictive model for Emergency Severity Index (ESI) triage classification is deemed feasible and clinically actionable. Analysis of 51,956 patient encounters reveals that routine triage data including vital signs, demographic variables, and chief complaint flags contains meaningful signals capable of discriminating between acuity levels. Successful implementation will require addressing class imbalance, external validation for target populations, and careful feature engineering.

Dataset Summary

The analysis is based on 51,956 patient encounters from a U.S.-based emergency department. The dataset includes structured variables across several domains: demographic information (age, race, ethnicity, gender, language, religion, marital status, employment status, insurance), administrative details (department name, arrival mode, arrival month, arrival day, arrival hour bin), vitals (heart rate, blood pressure, respiratory rate, oxygen saturation, temperature, glucose), and 200 chief complaint flags (binary indicators for presenting symptoms). The target variable is ESI, a five-level triage acuity scale where 1 represents the most urgent and 5 the least urgent.

Key Clinical Findings

ESI Class Balance
The ESI distribution reveals a significant class imbalance, with the majority of patients concentrated in the mid-range acuity classes (ESI 3-4). Higher acuity patients (ESI 1-2) and lower acuity patients (ESI 5) are substantially underrepresented. This imbalance will require mitigation during model training to prevent bias toward the majority classes and ensure adequate performance for high-risk patient identification.

Age Distribution
The patient population spans a wide age range, with a roughly normal distribution peaking in middle adulthood and extending into geriatric populations. Age shows meaningful correlation with triage acuity and should be included as a core feature in the final model.

Race and Ethnicity Distribution
The sample is predominantly White/Caucasian (approximately 25,000 patients) and Non-Hispanic (approximately 40,000 patients). Black/African American patients represent the second largest racial group, while other racial and ethnic categories are comparatively small. This distribution pattern raises important questions about external validity when generalizing to international or Caribbean settings where demographic composition differs markedly.

Chief Complaint Analysis

The top 15 chief complaints reveal that chest pain, shortness of breath, abdominal pain, and headache are the most frequently recorded presenting symptoms. Chest pain emerges as the single most predictive complaint for acuity classification, consistent with clinical expectations. The 200 complaint flags collectively provide substantial discriminatory power.

Vitals by ESI Class
Box plot analysis across ESI levels reveals differential patterns in vital sign distribution:

Heart Rate: Shows a clear gradient across ESI classes, with higher acuity patients exhibiting more extreme values. This is one of the strongest discriminators.

Respiratory Rate: Demonstrates a moderate separation across acuity levels.

Oxygen Saturation: Shows a strong gradient, with lower saturation values concentrated in higher acuity classes.

Blood Pressure: Provides moderate discrimination, with systolic blood pressure showing some separation.

Temperature: Exhibits weaker separation, suggesting less utility for acuity prediction.

Glucose: Shows minimal separation, with substantial overlap across classes.

Correlation Analysis
The correlation matrix confirms clinical expectations:

Heart rate and respiratory rate show the strongest correlations with ESI (-0.09 and -0.10, respectively)

Oxygen saturation correlates positively with ESI (0.18), meaning lower saturation predicts higher acuity

Blood pressure components show modest correlations

Temperature and glucose demonstrate minimal predictive signal

Top 3 Clinical Concerns & Mitigations

1. ESI Class Imbalance
Concern: The significant skew toward mid-range acuity classes may bias the model toward predicting majority classes, potentially underperforming for high-risk patients.

Mitigation: Apply stratified sampling during training, incorporate class-weighted loss functions, or use SMOTE with emphasis on minority classes. Implement cost-sensitive learning where misclassifying ESI 1-2 patients carries substantially higher penalty.

2. Data Representativeness & External Validity

Concern: The sample's demographic homogeneity—predominantly White and Non-Hispanic—limits generalizability to international or Caribbean settings where patient populations differ meaningfully.

Mitigation: Mandate prospective external validation using target-site data. Consider transportability techniques or domain adaptation. Develop site-specific calibration protocols.

3. Chief Complaint Feature Granularity
Concern: The 200 binary chief complaint flags, while powerful, may overwhelm the objective physiological signal. This could reduce model stability and generalizability.

Mitigation: Group complaints into higher-level semantic categories (cardiac, respiratory, trauma, neurological, gastrointestinal). Consider dimensionality reduction techniques to identify underlying complaint structures.

Top 3 Reasons for Feasibility

1. Strong Physiological Separation
Vital signs demonstrate clinically expected and statistically meaningful separation across ESI classes. Heart rate, respiratory rate, and oxygen saturation in particular show clear gradients, confirming that objective physiological data already contains robust predictive signal. The correlation patterns align with clinical intuition—more abnormal vitals predict higher acuity.

2. Data Quality & Completeness
Post-cleaning data shows minimal missingness in structured variables. Vitals were imputed by median imputation, preserving data integrity. Categorical variables were systematically coded with "Unknown" categories. This data quality supports reliable model development.

3. Clinically Coherent Feature Relationships
All observed correlations between features and ESI are directionally consistent with clinical expectations. The strongest predictors are physiologically meaningful (e.g., oxygen saturation, heart rate), and the chief complaint distribution reflects clinical triage priorities. This coherence supports construct validity.

Clinical Caveats & Limitations
External Validity
This model is derived from a U.S.-based sample and is not directly generalizable to Caribbean or international ED settings without careful recalibration. Differences in patient demographics, baseline health status, healthcare-seeking behavior, and resource availability may significantly affect model performance. External validation is essential prior to deployment.

Temporal Stability
Chief complaint patterns and vital sign norms are context-dependent and may drift over time. The model will require periodic retraining (e.g., annually) and ongoing performance monitoring to maintain accuracy.

Feature Exclusions
All leakage variables—including disposition and previous admission status—have been explicitly excluded from the feature set. This ensures the model is operationally sound for real-time triage use.

Top 10 Feature Shortlist (Leakage-Free)
Vitals:

triage_vital_hr — Heart rate (bpm)

triage_vital_sbp — Systolic blood pressure (mmHg)

triage_vital_dbp — Diastolic blood pressure (mmHg)

triage_vital_rr — Respiratory rate (/min)

triage_vital_o2 — Oxygen saturation (%)

triage_vital_temp — Temperature (°F)

triage_glucose — Blood glucose (mg/dL)

Demographics:

age — Patient age (years)

Administrative:

arrivalmode — Mode of arrival (walk-in, ambulance, etc.)

Chief Complaint:

cc_chestpain — Chest pain flag (top complaint with strongest acuity correlation)